



CPBRD Policy Brief

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AN EMPIRICAL ANALYSIS OF THE FACTORS AFFECTING REAL PROPERTY TAX COLLECTION IN THE PHILIPPINE CITIES

ABSTRACT

Real property tax or RPT provides the most stable own-source revenue for local governments and an important resource to improve service delivery and finance infrastructure projects (ADB, 2017). However, data shows that this source of income for local governments has not been fully maximized. The Department of Finance (2019) reported that local governments are losing P30.5 billion in revenues to outdated real property valuation alone. In particular, cities could have collected as much as P23.1 billion in incremental revenues from real property taxes had their schedule of market values (SMVs) been updated and in sync with international standards.

Reforms have been proposed by concerned government agencies to address the low revenue collection performance of RPT. At the House of Representatives, House Bill No. 6558 or the Real Property Valuation and Assessment Reform Act (Package 3) has already been passed on Third Reading in December 2022 in an effort to reform the country's valuation system.

This policy brief examines the factors affecting the RPT collections across cities in the Philippines using an annual dataset from the period of 2017 to 2021. The study employs three (3) panel regression models: the Pooled Ordinary Least Square, Random Effects, and the Fixed Effects Models, to analyze the significance and relationships of the factors or variables affecting RPT collection. The policy brief finds that among the factors examined, the *outdated real property valuation* as measured by the *number of overdue revisions* in the *Schedule of Market Value (SMV)* correlates significantly and negatively with RPT collections. This means that non-revision of SMV impacts RPT collections negatively. Furthermore, results show that for every 1-unit decrease in the score of overdue SMV general revisions is associated with an 18.90% increase in RPT collection or an estimated incremental revenue of P25.9 billion, controlling for internal revenue allotment, size of local economy index, and government efficiency index.

The views, opinions, and interpretations in this report do not reflect the perspectives of the House of Representatives as an institution or its individual Members.

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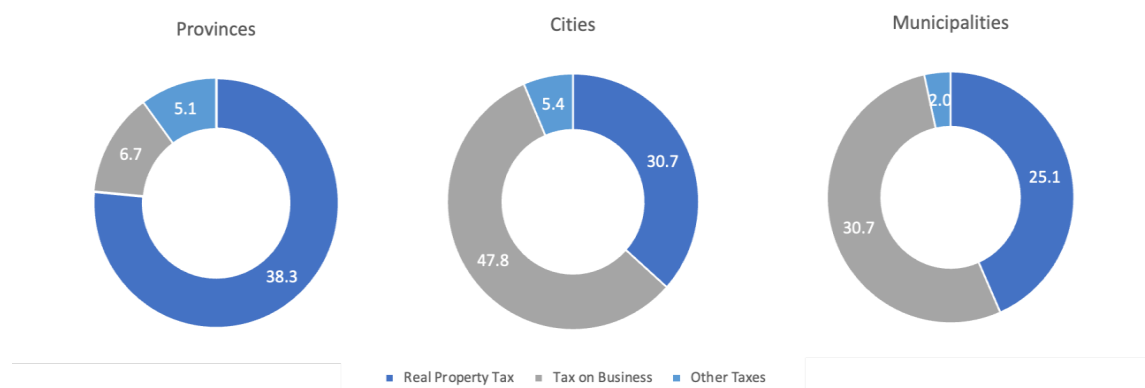
by JHOANNE E. AQUINO *

I. INTRODUCTION

Pursuant to Section 129, Chapter 1, Book II of Republic Act No. 7160 otherwise known as the Local Government Code (LGC) of the Philippines, the provinces, cities, and municipalities have the power to generate their own sources of revenues. These local sources consist of: (a) tax revenues derived from real property tax (RPT), business tax, and other taxes; and (b) non-tax revenues derived from fees, charges, and receipts from the operation of economic enterprises in connection with the government's exercise of its proprietary functions.

The real property tax or RPT collection is a major resource and potentially the biggest own-source revenues for local government units (LGUs) (ADB, 2020: BLGF, 2022). However, as seen in Figure 1, revenue performance varies greatly by LGU type. In 2021 alone, the bulk of local revenue sources of provinces, comprising 38.3% of the total, depends mainly on RPT. On the other hand, cities and municipalities predominantly obtained their local revenue sources from business tax collections yielding 47.8%. The RPT collections in cities stood only at 30.7% of total local revenue sources, and 25.1% in municipalities.

FIGURE 1
REVENUE PERFORMANCE, AS A PERCENTAGE TO TOTAL LOCAL REVENUE SOURCES,
BY LGU TYPE, 2021



Source of basic data: Bureau of Local Government Finance (BLGF), 2021

Revenues from real property constitute the most stable source of income for local governments and are important sources of resources to improve service delivery and finance infrastructure projects (ADB, 2017). It is estimated that land and real estate account for 45% to 75% of the wealth of developing countries (Ibotson, et.al., 1985 as cited by World Bank, 2017).

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In 2021, the construction industry contributed 11.1% to GDP growth, while in 2019, there was a 2.3% expansion in renting and business activities and a 3.4% increase in ownership of dwellings (BLGF, 2022). However, gains from these economic activities should have been translated into higher RPT collections. Note that the RPT collection performance from 2017 to 2019 has been minuscule as seen in its contribution to GDP growth at an average of 0.4% for the period 2017-2021. Likewise, the share of RPT collection to local revenue sources grew only by 0.5 percentage point from 30.0% in 2017 to 30.5% in 2021, averaging 29.0% in the same period (BLGF, 2022).

The literature attributes the low revenue collection of RPT to several factors. One is the outdated schedule of market values (SMVs) due to poor compliance with the LGC mandate to conduct a general revision of real property assessments. Based on Section 212 of the LGC, a schedule of fair market values shall be prepared by the provincial, city, and municipal assessors of the municipalities within the Metropolitan Manila Area for the different classes of real property situated in their respective LGUs for enactment by ordinance of the *Sanggunian* concerned.

Second is the lack of unified national standards and methods for valuation which has led to diversified and often conflicting valuation methodologies resulting in distorted real property values (Salceda, 2022; ADB, 2020). Various government agencies are requiring property valuation with each agency using its own system and methodology for valuation. The varying valuation and assessment methods of various LGUs have resulted in inequitable, disparate, confusing, conflicting, and often graft-prone valuation systems (Salceda, 2022). Adding to the complexity are the incomplete LGU assessment roles, which are overly dependent on the self-reporting of property owners about the values of their lands. Information about land and properties is also not frequently shared across government agencies and LGUs (World Bank, 2017).

Other factors attributable to the low revenue collection of RPT include: the susceptibility to tax evasion by unscrupulous taxpayers; and the absence of a complete and updated roll of real properties contained in a computerized database that truly reflects an accurate and current valuation of property listing (Salceda, 2022). ADB (2020) also noted that limited access to critical data and sparse use of information technologies have resulted in distorted real property values.

In this context, this policy brief seeks to provide an empirical analysis of the factors affecting the RPT collections across cities in the Philippines. This will first provide an overview of the RPT in the Philippines, followed by a review of related literature to examine the extent to which the identified factors or variables may affect and influence RPT collection.

For the succeeding sections, the paper will present the methodology and estimation results from the three (3) panel regression models: the pooled Ordinary Least Square, Random Effects model, and Fixed Effects model to examine the significance and relationships of the factors affecting RPT collection at the city level. This will be followed by a brief discussion on

the salient provisions of House Bill No. 6558¹ or the Real Property Valuation and Assessment Reform Act (Package 3), as well as the government agencies' efforts to reform the country's valuation system. Finally, the paper will discuss policy recommendations resulting from the findings of the empirical analysis. It is also hoped that these findings will serve as relevant reference material for legislators in the deliberations of related pending legislative measures.

II. Overview of Real Property Tax in the Philippines

The real property tax or RPT² is an annual recurrent ad valorem tax on real properties such as lands, buildings and improvements, and machineries. Section 233 of the LGC provides the ceiling rates applicable to respective localities:

- (a) In the case of a province at the rate not exceeding 1% of the assessed value of real property; and
- (b) In the case of a city outside or within the Metropolitan Manila Area or a municipality within the Metropolitan Manila Area, at the rate not exceeding 2%.

It is based on the assessed value of the property, which is derived from the application of the assessment levels to the fair market value³ of the property. The assessment level is the percentage applied to the fair market value to determine the taxable value of the property. (Title II, Chapter 1, Section 198 of LGC). The assessment levels are fixed by the ordinance of the local *Sanggunian* depending on the actual use of the property (National Tax Research Center (NTRC), 2016).

Section 218 of the LGC provides the following assessment levels:

On Lands. The assessment levels for land are differentiated depending on their actual use. Commercial, industrial, and mineral lands are assessed at the maximum rate of 50%. The assessment level for agricultural lands is at 40%, and residential and timberlands are at the lowest at 20%.

On Buildings and Other Improvements. For buildings and other improvements, the assessment levels are graduated, i.e., they increase directly with the fair market values, from 0% for residential buildings worth P175,000.00 and less to a maximum of 80% for commercial and industrial properties worth over P10,000,000.00.

¹ Approved by the House on Second Reading on 07 December 2022; approved on Third Reading on 12 December 2022; transmitted to the Senate on 12 December 2022; and received by the Senate on 14 December 2022.

² Section 247 of the LGC states that the collection of the real property tax with interest thereon and related expenses, and the enforcement of the remedies provided for in this Title or any applicable laws, shall be the responsibility of the city or municipal treasurer concerned. The city or municipal treasurer may deputize the Barangay treasurer to collect all taxes on real property located in the Barangay: Provided, That the Barangay treasurer is properly bonded for the purpose: Provided, further, That the premium on the bond shall be paid by the city or municipal government concerned.

³ Refers to the price of the property in an open or competitive market, wherein buyer is willing to buy and the seller is willing to sell on a given mutually agreed price.

On Machineries. Machineries are assessed in relation to their actual use. The lowest maximum level is assigned to machineries used for agricultural purposes at 40%, and the highest maximum level to commercial and industrial machineries at 80%. Machineries used for residential purposes are subject to a maximum assessment level of 50% (Section 218 of the LGC 1991: NTRC, 2019).

The tax due is computed by applying the basic RPT rates to the assessed value of the property (Section 233, LGC 1991: NTRC, 2019). Based on the LGC, the LGUs are given the opportunity to either lower or increase the assessment level and/or assessment rate or change both at the same time. To wit:

$$\text{RPT} = \text{Assessed value} \times \text{RPT rate}$$

Where,

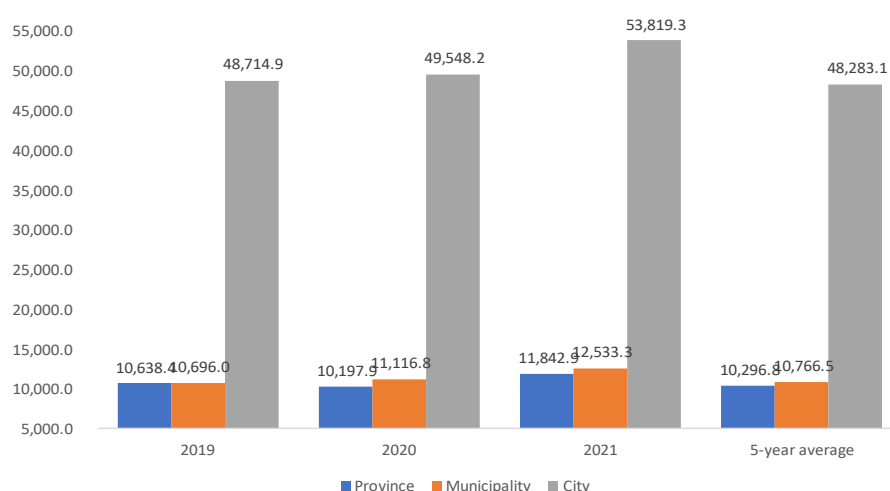
Assessed value Denotes the fair market value x assessment level;

RPT rate Denotes the rate (i.e., not exceeding 1% for provinces or 2% for cities/municipalities).

Real Property Tax Collection Performance

Figure 2 shows that among the LGUs, the cities comprise the largest share in total RPT collection⁴ averaging at P48.3 billion across a five-year period compared to provinces (P10.3 billion) and municipalities (P10.8 billion). Despite the imposition of the government's strict enhanced community quarantine in 2020 as an immediate response to the COVID-19 pandemic, the RPT collection in the cities still grew by 8.6% from P49.5 billion to P53.8 billion.

FIGURE 2
RPT COLLECTIONS, BY LGU TYPE, (IN MILLION PESOS)

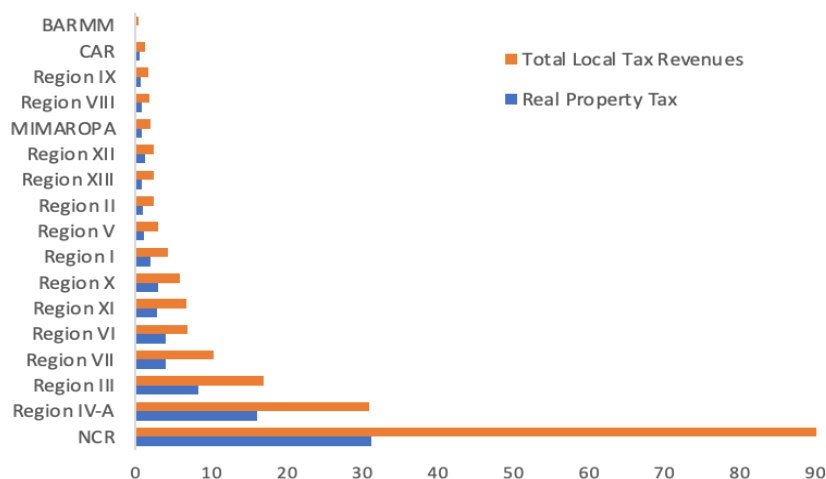


Source of basic data: BLGF (2019-2021)

⁴ Section 271 of the LGC mandates that the proceeds of the basic RPT shall be distributed as follows: a) in the case of Provinces: 35% shall accrue to the general fund of the province, 40% to the general fund of the municipality, and 25% to the Barangay; b) in the case of cities: 70% shall accrue to the general fund of the city, and 30% to be distributed among the component Barangays; c) in the case of municipality within Metro Manila: 35% shall accrue to the general fund of Metro Manila Authority, 35% to the general fund of municipality, 30% to be distributed among the component Barangays of the municipality.

By region, RPT collection in National Capital Region (NCR) generated the highest outtake in 2021, comprising P31.18 billion or 39.9% share to total RPT collection in the country (*Figure 3*). This is followed by Region IV-A with P16.1 billion (or 20.6% share) and Region III with P8.31 billion (or 10.6% share) RPT collections.

FIGURE 3
REAL PROPERTY TAX REVENUES VS. LOCAL TAX REVENUES, BY REGION,
(IN BILLION PESOS), 2021



Source of basic data: BLGF (2019-2021)

TABLE 1
REAL PROPERTY TAX COLLECTION PERFORMANCE, BY CITIES,
(IN BILLION PESOS), 2019-2021

City	2019	2020	2021	Average Share (%)	Average Growth (%)
Manila	5.2	4.6	5.4	10.0	4.5
Makati City	4.5	4.6	4.7	9.1	3.4
Quezon City	3.8	4.8	4.2	8.4	6.8
Pasay City	2.8	2.9	3.0	5.7	7.6
Taguig City	2.3	2.2	3.1	5.0	17.3
Parañaque City	1.6	2.1	2.2	3.9	7.4
Valenzuela City	1.3	1.5	1.8	3.0	9.7
Davao City	1.3	1.6	1.7	3.0	11.9
Mandaluyong City	1.3	1.3	1.5	2.7	7.2
Calamba City	1.0	1.1	1.4	2.3	14.9
Iloilo City	1.2	1.1	1.1	2.3	6.1
Muntinlupa City	0.9	1.1	1.1	2.0	10.5
Santa Rosa City	1.1	1.1	1.1	2.1	5.9
Pasig City	1.4	1.0	1.0	2.2	-20.7
Cebu City	1.0	1.0	0.9	1.9	1.8
Other 131 Cities	17.9	17.7	19.7	36.3	5.4
Total	48.7	49.6	53.8	100.0	5.4

Source of basic data: BLGF (2019-2021)

Looking into cities, RPT collections by the 146 cities nationwide consistently increased from P48.7 billion in 2019 to P53.8 billion in 2021, translating to a three-year average growth rate of 5.4%. The top 15 cities comprised 63.7% share of total RPT collections in 2021, 10 of which are dominated by cities from NCR (*Table 1*). A significant drop in RPT collection was observed in Pasig City for the period 2019-2021, declining from P1.4 billion in 2019 to P1.0 billion in 2021. Meanwhile, Quezon and Cebu Cities also posted a decline in RPT collection from 2020 to 2021.

Compared with the select ASEAN economies (*Table 2*), the average annual growth rate in the recurrent taxes on immovable property as a percentage of the total tax revenue of the Philippines is recorded at 2.1% in 2020, lower than Singapore's 6.7%. Lagging behind the Philippines are Indonesia (1.2%) and Thailand (1.1%). Immovable properties include land, buildings, road, and constructions of all kinds adhered to the soil.

TABLE 2
RECURRENT TAXES ON IMMOVABLE PROPERTY, SELECT ASEAN COUNTRIES
AS PERCENTAGE OF TOTAL TAX REVENUE, 2016-2020

Country	2016	2017	2018	2019	2020	Average
Indonesia	1.3	1.1	1.1	1.2	1.3	1.2
Lao PDR	0.7	0.7	1.1	1.2	1.2	1.0
Philippines	2.0	2.2	2.1	2.0	2.2	2.1
Singapore ⁵	7.4	6.7	7.0	7.0	5.1	6.7
Thailand	1.3	1.4	1.3	1.3	0.2	1.1
Vietnam	0.1	0.1	0.1	0.1	0.1	0.1

Source of basic data: OECD

III. Factors that may Affect the RPT Collection in Philippine Cities

Number of Overdue Revisions in Schedule of Market Values (SMVs)

Outdated SMVs adversely impacted the revenue performance of the RPT collection, which stands at 0.36% of GDP as of 2020 (ADB, 2020). In the last general revision of the BLGF in 2008, only 28% of the cities revised their SMVs (Salceda, 2022). Based on the data from the

⁵ In Singapore practice, revaluations take place every year (i.e., Annual Value) and the county has kept its effective property tax rates low as a matter of government policy. (ADB, 2022). Property tax on residential properties is found to be progressive. A progressive tax regime on residential property is equitable, as taxpayers with residential properties of higher value pay higher property tax. The tax rates for owner-occupied residential properties are lower than those for non-owner-occupied residential properties with the same Annual Value (MOF Singapore, n.d).

BLGF as of December 2021, evidence shows that more than half of the total number of cities, or 65% (96 out of 146) have outdated SMVs. In particular, there are cities within and outside Metro Manila that have failed to update their SMVs and conduct a general revision of property assessment for almost 27 years, which is already nine (9) times overdue. Based on the report of the BLGF in 2021, Table 3 shows the number of overdue revisions in the SMV of selected cities based on the years the SMVs were revised and approved for implementation.

TABLE 3
SMV COMPLIANCE, SELECTED CITIES

City	Implementation Year of the Approved SMV	No. of General Revisions Due
Cities in Metro Manila		
Makati, Parañaque, Pasig	1997	8
Mandaluyong	2001	7
Marikina	2002	6
Caloocan	2005	5
Taguig	2009	4
Cities outside Metro Manila		
San Fernando LU, Tuguegarao, General Santos	1994	9
Toledo	1998	8
Malolos, Guihulngan, Tanjay, Calbayog, Kidapawan	1999	7
Naga-Cebu, San Carlos, Lucena	2002	6
Batac, Lipa, Iloilo, Cebu City, Catbalogan	2006	5
Santiago, Tagbilaran, Cagayan de Oro, El Salvador	2008	4

Source of basic data: BLGF (2021)

It can also be observed that the base year used for the property valuation is far more outdated for some cities, resulting in higher estimated age of the property values. In fact, some cities within and outside Metro Manila are also using outdated property values. In particular, the estimated age of property values in Makati is 27 years behind (1994 as base year), while seven (7) cities outside Metro Manila were outdated by 20 to 31 years (*Table 4*). Since the SMVs are supposed to be updated every three years, the number of overdue revisions in these cities range from six (6) to ten (10) times.

TABLE 4
BASE YEAR OF PROPERTY VALUATION AND IMPLEMENTATION YEAR, SELECTED CITIES*

City	Base Year of Property Valuation	Implementation Year of the Approved SMV	Estimated Age of Property Values
Cities in Metro Manila			
Makati	1994-1999-2000	1997	27-22-21
Pasig	1996	1997	25
Marikina	2001	2002	20
Caloocan	2003	2005	18
Cities outside Metro Manila			
Pangasinan	1997-2001	2003	24-20
Cagayan	1990	1994	31
La Union	1993	1994	28
South Cotabato	1994	1995	27
Pampanga	1995	2014	26
Quezon	1995	2001	26
Cebu	1995	1998	26

**With 18 years and above estimated age of property values
Source of basic data: BLGF (2021)*

Based on a number of studies, the non-revision of SMVs of real properties has a significant negative influence on RPT collection. Property valuation is the backbone of real property taxation and regular valuation is vital to any property tax system to ensure correct taxes are collected and that fairness and equity are maintained (Pangcog, 1996). Hence, the failure of LGUs to update market values of properties greatly affects RPT collection, as well as negatively impacts the equity of the tax burden as property values do not reflect major physical and economic changes which have occurred in the locality (NTRC, 2014; ADB, 2020; Uy, 2021). In addition, the lack of technically qualified staff and assessment tools also impacts the assessment of real property (Bird and Slack, 2002).

There is limited research that used quantitative measures on the effectiveness of tax administration at a city level, and tax administration is in itself a broad concept (Das-Guta et. al., 2016 as cited by Uy, 2021). In this regard, the number of overdue revisions in the SMV⁶ is employed as a proxy variable to measure the effectiveness of tax administration, particularly having an outdated real property valuation. This proxy variable measures the number of times the city's SMV is overdue with respect to the requirement of the LGC.

⁶ The number of overdue revisions in SMV is computed by subtracting the year that the LGU last revised its SMV to the year of interest (i.e., 2017, 2018, 2019, 2020, 2021) and dividing the computed difference by three. An updated SMV will have a score of zero (Uy, 2021).

Undertaking valuation is important to enhance the revenue of a country. Updating property valuation methods as the basis for assessing property taxes would be able to structurally increase the property tax collections of the LGUs on a recurring basis, in view of the increase in property values over many decades (Tsala, 2021; Ricarfort, 2022 as cited by Jocson, 2022). Section 219 of the LGC requires LGUs to undertake a general revision of real property assessments within two (2) years after the effectivity of the LGC and every three (3) years thereafter.

Government Efficiency Index

Governance encompasses government, authority, control, and power. It includes the state's institutions and structures, decision-making processes, capacity to implement, and the relationship between government officials and the public (Landell-Mills and Serageldin 1992 as cited in Manasan, et.al. 1999). Given the complexity of the governance concept, this study employs Government Efficiency Index as a proxy variable to determine its influence with respect to RPT collection.

The Government Efficiency Index is one of the main pillars of the National Competitiveness Council's (NCC) Cities and Municipalities Competitiveness Index (CMCI)⁷. This index measures the quality and reliability of government services and government support for effective and sustainable productive expansion. Conceptually, this factor looks at government as an institution that is generally not corrupt, able to protect and enforce contracts, apply moderate and reasonable taxation, and is able to regulate proactively (La Porta et al, 1999 as cited in CMCI). As Salceda (2022) posited, the current valuation system of the country is susceptible to tax evasion by unscrupulous taxpayers who, through misdeclaration, misclassification, and other underhanded means, either diminish or increase the value of the real property, resulting in lower tax collection.

Uy (2021) revealed that for every 1% increase in the government efficiency index, the RPT collection is increased by 0.48%. This result is consistent with the literature that states that government efficiency has a positive influence on revenue collection. It is noteworthy that a pilot project supported by the Asian Development Bank (ADB) has significantly improved local tax revenue collection in Indonesia, increasing from 1.1% in 2012 to 1.3% in 2018 by modernizing local government taxation and policy improvement (BIMP-EAGA, 2022). Several countries in central Asia have a property-tax-to-GDP ratio higher than other countries in developing Asia partly attributable to having a system of centralized administration. This further supports the argument that improving tax administration and appropriate tax policy could result in increased revenues (McCluskey et.al, 2022).

⁷ Cities and municipalities are ranked on their competitiveness based on an overall competitiveness score. The overall competitiveness score is the sum of scores on five main pillars which pool data from several sub-indicators. The five main pillars are: economic dynamism, government efficiency, infrastructure, resiliency, and innovation.

Internal Revenue Allotment

The internal revenue allotment (IRA) is an intergovernmental fiscal transfer in the Philippines which is a substantial source of revenue for LGUs (Llanto, G. 2009). Under Section 248 of the LGC of 1991, LGUs are entitled to a 40% share of all national internal revenue taxes. However, following the Supreme Court (SC) Ruling on the Mandanas-Garcia Case in April 2019, the resources for LGUs increased as a result of a larger base for the computation of the just share of LGUs. In the first year of implementation of the SC ruling in 2022, the IRA now renamed as “National Tax Allotment or NTA” increased by P185.24 billion to P959.04 billion or about 19.10% of the 2022 national budget (Caraballo, 2021). With the phrase “internal revenue” deemed unconstitutional, the nomenclature for IRA has been changed to NTA beginning FY 2022 General Appropriations Act (GAA). For the share of LGUs in the NTA, the share remained the same as follows: 23% for provinces; 23% for cities; 34% for municipalities; and 20% for barangays. Meanwhile, the individual shares of LGUs shall still be determined based on population (50%), land area (25%), and equal sharing (25%) (DBM, 2023). Conversely, since the period of study (i.e., 2017-2021) only captures the implementation years of IRA, the term IRA shall be used instead of NTA.

Sections 18 and 286 of RA No. 7160 and Articles 383 and 390 of its Implementing Rules and Regulations specify that the share of LGUs shall be automatically and directly released to the provincial, city, municipal or barangay treasurer without the need for any further action, and shall not be subject to any holdback that may be imposed by the National Government (DBM, 2023).

There have been contrasting views in the literature on the effect of IRA on RPT collections. For instance, a number of studies show that LGUs, instead of becoming independent entities, especially in terms of revenue generation, became dependent on IRA (NTRC, n.d.). In particular, Manasan (1995, 2004, 2007) believed that from 1992 to 2000, IRA was being substituted for local tax revenues. NTRC (n.d.) noted that LGUs still rely heavily on IRA despite its revenue-raising powers. Accordingly, the share of IRA to total local revenues of the selected cities covered in this policy brief increased from 40.9% in 2019 to 44.7% in 2021, or with an annual average share of 42.1% (Table 5).

TABLE 5
INTERNAL REVENUE ALLOTMENT SHARE VS. REAL PROPERTY TAX COLLECTION PERFORMANCE,
SELECTED PHILIPPINE CITIES, (IN BILLION PESOS), 2019-2021

Total Selected Cities	2019	2020	2021	Average
IRA Share	127.8	143.2	154.5	141.8
% Growth Rate	10.6	12.0	7.9	10.2
RPT Collection	48.5	49.2	53.4	50.4
% Growth Rate	5.9	1.4	8.5	5.3
RPT Share to Total Local Revenues (%)	15.5	14.0	15.5	15.0
IRA Share to Total Local Revenues (%)	40.9	40.8	44.7	42.1

Source of basic data: BLGF (2019-2021)

On the other hand, some studies contradict the assumption that IRA was being substituted for local taxes, and that it creates a disincentive for local governments to raise their own revenue (Bird and Smart, 2002; Diana, 2008 as cited by Uy, 2021). Capuno (2003) revealed that IRA stimulated the local revenue performance of the Philippine provinces and cities during the period 1990-1996. Another study that employs a simple correlation estimate on the annual cross-section data from 2001 to 2006, albeit Philippine provinces level, also confirmed that IRA does not have any disincentive effect on local revenue mobilization (Diana, 2006). Notably, the result of the study of Uy (2021) showed that the IRA share of an LGU has a positive influence on their RPT collection, and for every 1% increase in IRA, RPT increases by 1.65%.

Size of Local Economy

The most well-known and frequent measure of economic growth at the national level is the gross domestic product (GDP). Numerous studies have revealed different results on the empirical relationship between tax revenue and economic growth. For instance, some studies indicated a positive relationship between taxation and economic growth (Anyanwu, 1997; Ogbonna and Appah, 2012; Yaya, 2013; Akwe, 2014), while other studies showed a negative relationship (Saibu 2015; Gareth, 2000; Bonu and Pedro, 2009; Saima, et.al., 2014 as cited by Majarjan, 2018).

Specific to RPT collection, Diana (2008) as cited by Uy (2021) revealed a positive relationship between economic development and RPT collection, pointing out that 1st and 2nd class provinces in the Philippines have a higher revenue effort compared to 3rd, 4th, and 5th-class provinces. This is attributed to the greater ability of richer provinces in collecting taxes than their poorer counterparts. In contrast, Bahl (2002, as cited in Bird and Slack, 2002) and Uy (2021) argued that economic development does not necessarily translate to high RPT collections. Uy (2021) further explained that the tax base of RPT (i.e., land, capital, machinery) is relatively inelastic. In general, property values (and, consequently, real property taxes) respond more slowly to annual changes in economic activity than other aspects such as income (and income taxes). Moreover, real property reassessment occurs only on a periodic basis (i.e., every five or ten years), hence, the failure to reflect the precise and updated real property values at the time of taxation (i.e., annually).

The literature on measuring economic growth at the local level, in particular, at the city level is very limited. Since the paper focuses on the effect of local economic development on real property tax at the city level, this will alternatively look into the size of local economy (SLE) index which measures the economic development of a city. The SLE is a sub-indicator under the Economic Dynamism pillar of the CMCI. In particular, this indicator approximates the level of economic activity in the LGU, which, at the national level, is measured by GDP. At the local level, the proxies for local economic activity include gross sales, which can be a measure of local production, and the number of business registrants and total capitalization of newly registered business enterprises, which indicate the level of new investment in the locality.

IV. Data Used and Methodology

Data Used, Scope, and Limitations

The paper expands the scope of the previous study of Uy (2021) to include all the cities outside Metro Manila, rather than only those within the National Capital Region. This provides a larger representation of cities in all regions of the country. A panel data set which consists of annual observations for the period 2017 – 2021 for 139 cities out of the total 146 cities in the Philippines was collected for the study. Note that some cities were removed from the list of observations due to the unavailability of data (i.e., Canlaon, Lamitan, and Marawi Cities have no data on the size of local economy index; Isabela City has no government efficiency index data; Sto. Tomas, Talisay of Cebu City, and Island Garden City of Samal also lack relevant information across the period of study). The data obtained from the BLGF were: 1) RPT collections, 2) the amount of IRA/NTA, and 3) the number of overdue revisions in SMV. The size of the local economy index and government efficiency index were both culled from the NCC's CMCI. The CMCI is an annual ranking of Philippine cities and municipalities developed by the NCC.

TABLE 6
DESCRIPTION OF INDEPENDENT AND DEPENDENT VARIABLES

Variable	Indicator	Description	Expected Relationship to RPT Collection	Source
RPT collection	RPT collections of selected Philippine cities	An annual recurrent ad valorem tax on real properties based on set ceiling rates (1% for properties located in a province and 2% for properties located in a city or a municipality within the Metropolitan Manila Area). RPT = assessed value of the real property x RPT rate		BLGF
Outdated Real Property Valuation	Number of overdue revisions in SMV	Measures the effectiveness of tax administration at the city level and how up-to-date the city's valuation system is with respect to the requirement of the LGC. Calculated as follows: No. of SMV revisions due = (year of interest-year of last SMV revision)/3 An updated SMV will have a score of zero. The higher the score, the more outdated is the SMV of an LGU.	(-) (Pangcog, 1996; NTRC, 2014; ADB, 2020; Uy, 2021; Jocson, 2022)	BLGF
Governance	Government efficiency index	An index that measures the quality and reliability of government services and support for effective and sustainable productive expansion.	(+) (Uy, 2021, McCluskey, et.al, 2022)	NCC
Fiscal transfer	IRA	An intergovernmental fiscal transfer which is determined based on population (50%), land area (25%), and equal sharing (25%).	(+) (Capuno, 2003; Diana, 2006; Uy, 2021) (-) (NTRC, n.d., Manasan, 2007)	BLGF
Local economic development	Size of local economy index	An index that measures the economic development of a city through gross sales, the number of business registrants, and total capitalization of newly registered business enterprises.	(+) (Diana, 2008) (-) (Bahl, 2002; Bird and Slack, 2002)	NCC

Sources: Uy (2021), NCC (2023), DBM (2023)

The dependent variable in the paper is the RPT collections in nominal terms, while the independent variables are the potential factors that affect the RPT collections. The size of local economy is used as a proxy variable to measure the significance and relationship of economic development at the city level, while the IRA as the intergovernmental fiscal transfer. Various analyses have differing views on the effect of economic development on RPT collection and there are only limited in-depth analyses on the relationship at the city level. Hence, the findings of the paper will serve as additional research material to establish the relationship between the two (2) variables at the city level (*Table 6*).

The variable of interest is the number of overdue revisions in the SMV, which data is readily available and easy to quantify. As quantitative measure for proxy variables for the effectiveness of tax administration is scarce, this indicator is used to measure the outdated real property valuation at the city level. On the other hand, the government efficiency index is used as a proxy variable to measure governance at the city level and to determine the significance and relationship with respect to RPT collection.

Research Methodology

To determine the relationship and significance between the identified factors/independent variables (i.e., number of overdue revisions in the SMV, government efficiency index, IRA, economic development,) and dependent variable (i.e., RPT collection), the paper considers three (3) panel regression models as follows: **the Pooled Ordinary Least Square (OLS), Random Effects Model, and Fixed Effects (FE) Model**. This econometric analysis has been employed in other studies which are used to assess the association or model the relationship between two or more independent/explanatory variables and a single dependent/response variable or estimate the impact of various determinants on the property tax revenues (Awasthi, 2020; Yale University, n.d; Boston University, n.d. as cited by Uy, 2021).

Pooled Ordinary Least Square (OLS) happens to be a commonly considered model for panel datasets. However, this model is likely to cause endogeneity when the residuals are correlated with the independent variables which can lead to incorrect inferences about the relationships between the independent and dependent variables. To address the problem of endogeneity, alternative estimation techniques such as fixed effects and random effects will be used. **A Random Effects (RE)** model assumes the variation across entities to be random and uncorrelated with the independent variables included in the model. On the other hand, **Fixed Effects (FE)** model is an estimation technique employed in a panel data setting that allows researchers to control for time-invariant variables⁸ and estimate the parameters of interest while holding constant the individual-specific effects.

⁸ By time-invariant values means that the value of the variable does not change across time.

Pooled OLS:

$$RPT_{it} = \beta_0 + \beta_1 SMV_REV_{it} + \beta_2 \ln IRA_{it} + \beta_3 \ln SLE_{it} + \beta_4 \ln GE_{it} + \varepsilon_{it}$$

Where:

RPT_{it}	City's real property tax collection; i = city and t = year
β_i where $i=1,2,3,4$	Regression coefficients to be estimated
SMV_REV_{it}	Number of overdue revisions in SMV (measures the effectiveness of tax administration of a city/how up-to-date is the valuation system of a city)
SLE_{it}	Size of local economy index (measures the economic development of a city)
IRA_{it}	City's IRA share (measures fiscal transfers)
GE_{it}	Government efficiency index (measures the effectiveness of tax administration through the quality and reliability of government services and support)
ε_{it}	error term

Fixed Effects (FE):

$$RPT_{it} = \beta_0 + \beta_1 SMV_REV_{it} + \beta_2 \ln IRA_{it} + \beta_3 \ln SLE_{it} + \beta_4 \ln GE_{it} + \alpha_{it} + \varepsilon_{it}$$

Where:

α_{it} City fixed effects; $i = 1, 2, \dots, 139$

Assumption: $Cov(\alpha_i, X_{it}) \neq 0$

Random Effects (RE):

$$RPT_{it} = \beta_0 + \beta_1 SMV_REV_{it} + \beta_2 \ln IRA_{it} + \beta_3 \ln SLE_{it} + \beta_4 \ln GE_{it} + \mu_{it}$$

Where:

$$\mu_{it} = \alpha_i + \varepsilon_{it}$$

Assumption: $Cov(\alpha_i, X_{it}) = 0$

Before arriving at the final model to use, the first step is to determine whether the pooled OLS is more appropriate to use than the RE model. In choosing between the two (2) models, the Breusch-Pagan Lagrange Multiplier (LM) Test is performed. The null hypothesis is that the variance of the RE model is zero versus the alternative hypothesis that the variance of the RE model is non-zero. If the null hypothesis is rejected, then the RE model is more appropriate than the usual OLS regression model. By employing the Breush-Pagan LM Test, the result shows that the p-value is less than 5% level of significance (*Annex A*), hence, this study rejects the null hypothesis that the pooled OLS is appropriate. Therefore, the RE model will be the appropriate model to use.

Subsequently, in choosing between the RE and FE models, the Hausman Test is performed. The null hypothesis of the test is that the preferred model is the RE model while the alternative hypothesis is that the FE model is preferred. If the test is significant, then the FE model will be used. By using Hausman Test, the result shows that the p-value is less than 5% level of significance, hence, this study rejects the null hypothesis that RE model is appropriate (*Annex A*). Therefore, FE model will be the appropriate and final model to use in this study.

Note that there are other diagnostic tests for panel data such as tests for cross-sectional dependence and tests for serial correlation. However, these tests are applicable only for panel data with long time series. Considering that the number of panels or time series of this policy brief is only a few, such diagnostic tests were not applied. If there is evidence of non-constant variance, then robust standard errors will be used. Likewise, testing for the presence of multicollinearity among independent variables in a multiple regression model using variance inflation factor (VIF) is also not recommended under a fixed effects model, where you have dummy variables to control for time-invariant individual effects. This is because the dummy variables are perfectly correlated with each other within the same unit, resulting in higher VIFs. Note that the higher the VIF, the higher the possibility that multicollinearity exists.

Key Findings and Estimation Results

The regression results of the 139 cities are presented in Table 7. In this table, the paper compared the regression results between pooled OLS (Column 1) and FE (Column 3) models. In the pooled OLS model, the size of the economy as measured by the *size of local economy index* has a significant and positive effect on the real property tax revenue while *government efficiency index* has also a positive effect on real property tax revenue, albeit, statistically insignificant.

Meanwhile, the coefficient of the *number of overdue revisions* in SMV is insignificant. *IRA* on the other hand, has a positive effect on RPT, which is a bit counterintuitive. Such a result might be because when running pooled regression, all the observations are being pooled together for the estimation without controlling for time-invariant unobserved individual characteristics that can be correlated with the observed independent variables. This could cause potential omitted

variable bias due to unobserved heterogeneity⁹ when this heterogeneity is constant over time. Hence, to remove potential heterogeneity from the data, the FE model was employed.

TABLE 7
PANEL REGRESSION RESULTS ON THE EFFECT OF NON-REVISION OF SMV ON RPT^{10,11,12}

Variables	(1) Pooled OLS	(2) Random effects	(3) Fixed Effects ¹³
<i>SMV_REV</i>	-0.0172 (0.0139)	-0.0771*** (0.0233)	-0.1890*** (0.0602)
<i>lnIRA</i>	0.0867 (0.0623)	-0.0801 (0.0700)	-0.1015 (0.0808)
<i>lnSLE</i>	0.376*** (0.0220)	0.0750*** (0.0194)	0.0023 (0.0194)
<i>lnGE</i>	0.209 (0.180)	0.0231 (0.0929)	-0.0671 (0.1166)
Constant	1.917*** (0.575)	2.711*** (0.551)	2.662*** (0.800)
Observations	688	688	688
R-squared	0.391		0.95
Number of cityid		139	139

Standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Column 3 shows the regression results from the FE model. The size of local economy index although still positive now turns insignificant. Meanwhile, the *government efficiency index* now shows a negative effect on RPT collections, although insignificant. Earlier OLS regression results for both size of local economy and government efficiency index have positive effects on RPT collections because the variations across the cities were considered. But when FE model was used, the estimated impacts within cities, the effects from the *size of local economy and government efficiency* are minimal, since both variables within each city maintain very similar levels across time.

⁹ Defines as a dissimilarity between elements that comprise a whole. When heterogeneity is present, there is diversity in the characteristic under study (<https://statisticsbyjim.com/basics/heterogeneity/>).

¹⁰ For robustness check, see Annex B for the regression results where a dummy variable COVID2020 was added to capture the reduction in RPT collections due to the COVID-19 pandemic.

¹¹ Another robustness testing was performed running the same set of models but using RPT as a percentage of locally-sourced revenues as the outcome variable, see Annex C. Similar to the final model, under the FE model, the number of overdue revisions in SMV shows negative coefficients and are statistically significant at 10% level significance.

¹² Additional robustness check was performed using the dependency ratio (i.e., IRA to Total Revenue) variable instead of using the usual IRA in nominal values. Regression results under the FE model reflect that the dependency ratio shows positive coefficients which is counterintuitive and is statistically insignificant.

¹³ Applied cluster-robust standard errors to account for possible heteroscedasticity and serial correlation.

As an additional explanation, results from the RE regression are also presented in Column 2. There are some differences between RE and FE model models, in terms of magnitude and significance, in particular for the *number of overdue revisions in SMV, IRA, and size of local economy*. However, the signs are similar for all coefficients in both models. As explained above, it is possible that omitted variables have a correlation with the present variables. Furthermore, the result of the Hausman Test leads to indicate that the estimates generated by the FE model are more accurate.

Key findings of the paper also reveal notable changes in the coefficients of *IRA* and the *number of overdue revisions in SMV*. *IRA* has now a negative coefficient as predicted, albeit insignificant. Among the factors analyzed, the **number of overdue revisions in SMV shows a negative effect on RPT collections as expected, and is now statistically significant at 1% level of significance**. This suggests that a one (1) - unit decrease in the score of the number of overdue revisions in SMV is associated with an 18.90% increase in RPT collection. This also translates to an estimated incremental revenue amounting to P25.9 billion¹⁴.

This further supports the argument of Uy (2021) that the non-revision of SMVs of real properties has a significant negative influence on RPT collection. The study's findings are also consistent with the observations of NTRC (2014) and ADB (2020) that failure to update SMVs will greatly affect RPT collection as well as negatively impact the equity burden as property values do not reflect major physical and economic changes which have occurred in the locality. According to the DOF (2019), cities could have collected as much as P23.1 billion in incremental revenues from RPT collection had their SMVs been updated and in sync with international standards. The P23.1 billion RPT collection that cities fail to collect could have built either 513 transport terminals, 339 landfills, 1,154 satellite health centers, or 3,330 low-cost resettlement projects (BLGF, 2019).

V. Existing Tax Policy Framework and Salient Features of Proposed Tax Reform Measure

To improve the real property valuation system of the government and to support its long-term revenue efforts, the House of Representatives approved House Bill No. 6558 or Real Property Valuation and Assessment Reform Act on Third Reading last 12 December 2022. HB No. 6558 particularly seeks to promote a just, equitable, impartial, and nationally-consistent real property valuation based on internationally-accepted valuation standards and principles. The proposed bill, which is a consolidation of several measures filed in the House plenary, is considered one of the priority legislative measures of the current administration as mentioned in the President's first State of the Nation Address (SONA). The proposed measure is also considered as a priority measure in the Philippine Development Plan 2023-2028 to further support the capacity of local governments in resource generation.

¹⁴ Derived by multiplying the total 2021 RPT collection of the cities under observation by the average score of overdue revisions in SMVs of the total cities under observation and the given coefficient (18.9%).

Specifically, to address the issues and challenges identified in the previous section of this policy brief, some of the key provisions stated in the proposed measure are as follows:

- a) mandate that the SMVs shall be updated within two (2) years from the effectivity of the proposed measure;
- b) reorganize and strengthen the Bureau of Local Government Finance in order for it to effectively provide institutional support to the operationalization of the government's reform agenda;
- c) mandate the BLGF to lead and manage the implementation of the proposed measure, as such:
 - i. develop, adopt and maintain valuation standards consistent with generally accepted international valuation standards, regulations and specifications for real property appraisal use for taxation and other purposes;
 - ii. in coordination with the BIR, review and ensure that SMVs as prepared by the local government assessor are compliant with real property valuation policies and standards; and
 - iii. provide technical assistance to government agencies, leadership and policy direction to LGUs on real property valuation for taxation, uniform procedure on different transactions in the assessor's office, and conduct continuing study and research on valuation.
- d) require that the valuation or appraisal of all real property whether taxable or tax-exempt, shall be based on the prevailing market values in the locality where the property is situated in conformity with the valuation standards adopted under the proposed bill.
- e) mandate the creation of the Real Property Valuation Service within the BLGF with counterpart personnel in BLGF Regional Office.
- f) require the Secretary of Finance to approve the SMVs within sixty (60) days from receipt of the joint endorsement of the BLGF Executive Director and the Commissioner of Internal Revenue; and among others.
- g) include penal provisions on non-revision of SMVs.

A counterpart bill, Senate Bill No. 314 authored by Senator Juan Miguel "Migz" F. Zubiri, was filed on 12 July 2022 in the Senate and is pending with the Committees on Ways Means, Local Government, and Finance as of 08 August 2022.

Additionally, there are already existing reform efforts undertaken by the government to improve property tax collection efficiency at the local level. For instance, the DOF issued on 19 October 2009 the Department Order (DO) No. 37-09 Philippines Valuation Standards to raise the quality of public and private sector valuations, provide consistency and understanding between providers and users of valuations, and reduce financial risk for users of valuation.

DOF issued DO Order No. 10 or Mass Appraisal Guidebook for LGUs in 2010 to capacitate the assessors. In 2010, DOF-DILG Joint Memorandum Circular No. 2010-01 was issued to reiterate the requirement of the LGC for regular revaluation of SMV. On the other hand, the SMVs of all LGUs are now downloadable online following the issuance of BLGF Memorandum Circular No. 16-2011. Prototype versions of the Electronic Field Appraisal and Assessment Sheet (eFAAS) and Valuation Database and Information System (VDIS) are currently being enhanced by the government concerned. VDIS is a data storage facility for capturing sales transactions in the LGUs while eFAAS serves as a start-up computerized RPT administration and records management for low-income class LGUs to back up and manage all assessment records presently in manual form and to generate computer-based reports.

VI. Summary and Conclusion

Existing literature regards property tax as a perfect tax because land in particular being a capital asset is visible and immovable. This is an important fiscal tool in this time of globalization and competition (World Bank, 2017). Likewise, property taxes should be a key area of focus for developing economies in Southeast Asia seeking to raise government revenue after the pandemic (ADB, 2022). However, specific factors affecting RPT collections at the national or local level are relatively less researched and there are only limited studies that provide an empirical analysis of the extent to which such potential factors may affect RPT collections. Hence, this paper made an attempt to examine the significance and relationships of the potential factors affecting RPT collection in 139 cities in the Philippines across a five-year period from 2017 to 2021 and to further contribute as reference material for legislators in the deliberations of related pending legislative measures. This study utilizes three (3) panel regression models: pooled OLS; RE model and FE model. As expected, the FE model fits the data better due to potential endogeneity issues.

The study's findings provide empirical evidence that among the factors or variables analyzed, the proxy variable for the effectiveness of tax administration, particularly on outdated real property valuation —**the number of overdue revisions in SMV (variable of interest), correlates significantly and negatively with RPT collections**, that for every one (1)-unit decrease in the score of SMV general revisions due is associated with an 18.90% increase in RPT collection, translating to an estimated incremental revenue of P25.9 billion. The study's result further validates that poor compliance with the LGC mandate to conduct an SMV general revision impacts the RPT collections, particularly at the city level. The study's result further provides a strong argument that despite continuous development in cities, the RPT collection of the cities nationwide grew only at 5.3% from 2019 to 2021. When compared to the ASEAN countries, the recurrent taxes on immovable property as a percentage of total tax revenue of the Philippines only averaged 2.1% against Singapore's 6.7% across the years 2016 to 2020.

In this context, policy reforms such as strictly enforcing the updating of SMVs of real properties must be adhered to by the LGUs to improve RPT collections. The existing LGC has no corresponding sanctions that could compel LGUs to update their SMVs. However, Article IV of the Real Property Valuation and Assessment Reform (RPVAR) Act already includes penal

provisions for the failure to prepare the SMVs; failure to comply with valuation standards; government agents delaying the review, approval and implementation of SMV; and failure to conduct general revisions.

There is also a need to review options obliging LGUs to update SMVs based on market values such as the provision of incentives to those who will abide by the mandatory general revision (World Bank, 2017). If properly implemented, this will broaden the tax base, thereby increase revenue collections. In Naga City experience alone, a market-based revision of SMV yielded at least 27% increase in RPT collections of the LGU (BLGF, 2022). Accordingly, BLGF estimates that an incremental revenue of P18.6 billion to P37.5 billion (full implementation) is expected to be generated should the RPVAR Act be passed into law.

The *internal revenue allotment (IRA)* (measured by intergovernmental fiscal transfer based on population, land area, and equal sharing), *size of local economy index* (measured by the economic development of a city through gross sales, the number of business registrants, among others) and *government efficiency index* (measured by the quality and reliability of government services and support for effective and sustainable productive expansion) were found to be statistically insignificant when FE model was used. This suggests that the data coverage may need to be extended or adjusted to improve the significance of the results. It is important to note that, as previously discussed, the effects of these variables on the size of local economy and government efficiency may be minimal since they tend to remain at similar levels within each city over time.

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ANNEX A
SELECTION METHODS OF REGRESSION PANEL DATA:
POOLED OLS VS RANDOM EFFECTS AND FIXED EFFECTS

MODELS

Particular	Breusch and Pagan (LM) test (OLS vs RE)	Hausman test (FE vs RE)
chibar2	871.54	2572.54
Prob > chibar2	0.0000	0.0000

ANNEX B
PANEL REGRESSIONS RESULTS CONTROLLING
FOR THE COVID-19 PANDEMIC IN 2020

		effects	effects
<i>SMV_REV</i>	-0.0157 (0.0138)	-0.0621*** (0.0231)	-0.189*** (0.0602)
<i>lnIRA</i>	0.101 (0.0623)	-0.0109 (0.0699)	-0.102 (0.0808)
<i>lnSLE</i>	0.376*** (0.0219)	0.0734*** (0.0190)	0.00228 (0.0194)
<i>lnGE</i>	0.190 (0.179)	0.0531 (0.0910)	-0.0671 (0.117)
<i>COVID2020</i>	-0.192** (0.0748)	-0.144*** (0.0268)	-0.0635 (0.0533)
Constant	1.904*** (0.573)	2.170*** (0.550)	4.853*** (0.751)
Observations	688	688	688
R-squared	0.397		0.951
Number of cityid		139	

Standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

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